

nRQL Cheat Sheet the RacerPro Query Language · verified on RacerPro 2.0 · github.com/lambdamikel/shacl-nrql-comparison

Model: a positive atom (`?x C`) binds `?x` only where the KB *entails* it (open world, DL reasoning); **neg** is *negation as failure* over *known* objects (closed world). **No unique-name assumption.**

Server & protocol

```
LD_LIBRARY_PATH=. ./RacerPro ⇒ TCP 8088.
Send "cmd\n"; read CR-terminated frame:
:answer <id> "res" "out" / :ok / :error
```

KB setup

```
(full-reset)
(in-tbox NAME)           ; current TBox
(in-abox NAME [TBOX])    ; current ABox
(in-data-box NAME)       ; substrate
(in-mirror-data-box NAME) ; auto-mirror
```

TBox

```
(implies Dog Animal) ; Dog<=Animal
(equivalent Mother
  (and Woman (some has-child Human)))
(disjoint Man Woman)
(define-primitive-role has-child
  :inverse has-parent :transitive nil
  :domain Human :range Human)
(inverse withdrawal deposit)
(implies-role deposit transaction)
(define-concrete-domain-attribute
  age :type integer)
```

Concept expressions

```
(and ..) (or ..) (not C) (some R C) (all R C)
(at-least n R [C]) (at-most n R [C])
(exactly n R [C]) (one-of i ..)
(an attr) ; "has some concrete value"
top bottom
```

ABox

```
(instance alice Woman)
(related alice junior has-child)
(constrained alice a-age age)
(constraints (equal a-age 50)) ; int->equal
(different-from k1 k2) ; no UNA!
```

CD preds: int (equal a v), real (= a v)(> a v), string (string= a s).

Query forms

```
(retrieve <head> <body>)
(retrieve1 <body> <head>) ; for :reject
(tbox-retrieve <head> <body>)
(defquery NAME (vars) <body>)
```

Variables

```
?x ; ABox var, NON-injective
$x ; ABox var, INJECTIVE (distinct!)
?*x $?*x ; substrate node (non-inj / inj)
*ind ; a named substrate individual
```

NB: reverse of the UG (RacerPro 2.0 was changed; `$?` is injective). Force distinctness explicitly — see Gotchas.

Body atoms

```
(?x C) ; C=ANY concept expr (entailed)
(?x (some r top)) ; OWA: has some r-filler
(?x ?y R) ; role
(?x ?y (inv R)) ; inverse role
(?x (has-known-successor R))
(TOP ?x) (BOTTOM ?x)
(same-as ?x ?y) | (= ?x ?y)
```

Body constructors

```
(and B..) (union B..) (neg B) (inv B)
(project-to (?x..) B) ; B may be a conjunction
(substitute (NAME args...))
;; "c's with no r-successor of type e"
;; (complex projected body, negated):
(retrieve (?x) (and (?x c)
  (neg (project-to (?x)
    (and (?x ?y r) (?y e))))))
```

Head projection

```
(?x) ; just the bindings
(?x (told-value (age ?x))) ; a told value
;; aggregation via head lambdas (MiniLisp)
```

Killer idioms (live-verified)

```
;; sh:minCount 1 (= NOT EXISTS)
(retrieve (?x) (and (?x Person)
  (neg (project-to (?x) (?x ?y fn)))))
;; sh:class on a path (no countereample)
(retrieve (?x) (and (?x Parent)
  (?x ?y has-child) (neg (?y Person))))
;; sh:maxCount 1 (no UNA: over-reports)
(retrieve (?x) (and (?x Person)
  (?x $?a fn) (?x $?b fn)
  (neg (same-as $?a $?b))))
;; sh:disjoint (shared value)
(retrieve (?x)
  (and (?x ?v likes) (?x ?v dislikes)))
```

Data substrate / property graphs

```
(data-node acc1
  ((:amount 1000.0) (:currency $)) account)
(data-edge acc1 acc2
  ((:amount 9900.0)) deposit) ; edge props!
(node-label acc1) (edge-label acc1 acc2)
(retrieve (?*x)
```

```
(?*x ((:amount (:predicate (> 1e4)))))
(retrieve (?*x ?*y)
  (?*x ?*y (:predicate (equalp deposit))))
(?*x (:predicate (search "Joh"))) ; substr
```

Rules

```
(firerule
  (and (?x ?y has-child) (?y ?z has-child))
  ((related ?x ?z has-grandchild)))
; antecedent may use neg; consequent may
; (instance (new-ind tag ?x) C) -- create.
```

Reasoning (what a validator can't)

```
(classify-tbox) (taxonomy)
(concept-subsumes? C D) ; D<=C inferred
(concept-satisfiable? C)
(realize-abox)
(individual-instance? i C)
(individual-types i)
(concept-instances C)
(abox-consistent?)
```

Query reasoning (the exotic part)

```
(prepare-abox-query (?x) (?x Mother)) ; q-1
(query-consistent-p :query-1) ; ever nonempty?
(query-entails-p :query-1 :query-2) ; subsump
(query-equivalent-p :query-1 :query-2)
```

OWL

```
(owl-read-file "/path/onto.owl")
(concept-instances #:DogOwner) ; inferred
(individual-instance? #:john #:DogOwner)
```

MiniLisp hatch (termination-safe!)

```
(evaluate (+ 1 2)) ; restricted evaluator
;; retrieve1 head with a
;; (:lambda (x) .. :reject) = a filter
```

Has arith, format, search, subseq, reduce, dotimes, KB callbacks. **Lacks** recursion (aborted to ensure termination), loop, require, regex. Not Turing-complete — by design, for a decidable reasoner.

Gotchas

- **No UNA:** (at-most 1 R) + 2 named fillers is *consistent* until (different-from...).
- (neg (?x ?y R)) ≠ (neg (project-to (?x) (?x ?y R))) — use the 2nd for "no R-successor".
- **neg** = failure over *known* objects, not classical negation, not "triple absent".
- has-known-successor sees *named* successors; an OWL existential's anonymous witness it won't.
- **Var inequality:** (neg (same-as ?x ?y)) or (neg (= ?x ?y)). No <> for individuals; different-from is ABox-only. UNA toggling doesn't change query var-distinctness.